Rejection Sampling with the Ziggurat Method

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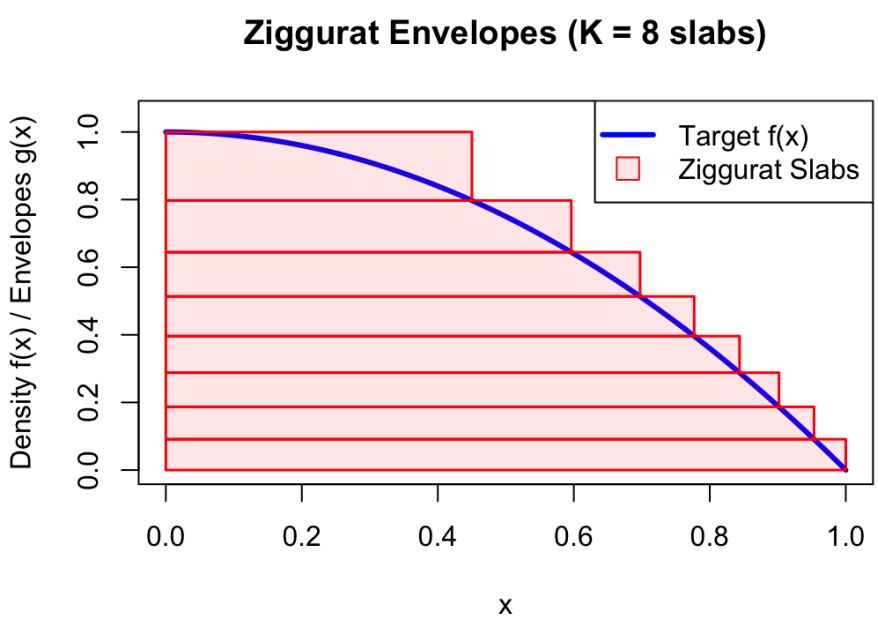
Introduction

The Ziggurat Method is a way of generating exact samples from a probability distribution. It is a specific instantiation of Rejection Sampling, and is typically applied to sampling from univariate distributions with a decreasing density. It is most straightforward to describe in the case of a compactly-supported density, and so we focus the main discussion on continuous, (strictly) decreasing (unnormalised) densities $f : [0, 1] \to [0, 1]$ with $f(0) = 1$ and $f(1) = 0$, with the understanding that the general case will follow relatively easily from this case.

Figure 1: The ziggurat envelope for $f(x) = 1 - x^2$ with $K = 8$ slabs



Rejection Sampling

The core notion which underpins Rejection Sampling is that if you can upper bound a density of interest by a more tractable density which admits straightforward sampling, then a simple selection strategy allows for exact sampling from that density of interest. In practice, developments in Rejection Sampling hence tend to arise from identifying tractable dominating families which are reasonably good approximations to densities of interest.

Balanced Histograms and Fast Sampling

Arguably the simplest densities to consider in this context would be histograms, i.e. piecewise-constant densities. Simulation from such distributions amounts to a combinatorial step (selecting which cell to sample from) and a uniform simulation (to sample within the corresponding interval). Note that the first step is particularly simple if the areas of all cells involved are equal (since uniform distributions over discrete sets indexed as $\{1, 2, \cdots, K\}$ can be simulated easily by rounding a standard $\mathsf{Uniform}\,[0,1]$ random variable), and becomes even simpler if the number of cells is a power of two (since if $K = 2^B$, one can simulate the associated index using only B fair bits, with no continuous random variables needed).

Ziggurat Envelopes for Rejection Sampling

The Ziggurat Method is then based on this principle: given a decreasing density f as before, and a budget $K = 2^B$, find an upper-bounding g which is made up of K 'slabs' of equal area, and fits 'as tightly as possible' around f.

The convention in the Ziggurat method is to construct g 'vertically', i.e. in the form

$$g(x) = \sum_{1 \le k \le K} \theta_k \cdot \mathbf{1}\left[0 \le x < w_k\right],$$

with $1 = w_1 > w_2 > \cdots > w_K > 0$, whence the name 'ziggurat'[1].

Algorithm 1 Sampling from a Ziggurat Representation

1. Sample $I \sim \mathsf{Categorical}\left((\theta_1, \theta_2, \cdots, \theta_K)\right)$.
2. Given $I = k$, sample $X \sim \mathsf{Uniform}\,[0, w_k]$.

Among other things, this makes the 'equal-area' constraint easier to manage by enforcing that the area of each slab (obtained by multiplying 'thickness' θ_k and the 'width' w_k) be independent of k. One can of course refactor this density 'horizontally' by observing that with 'height' variables $h_k := \sum_{1 \le \ell \le k} \theta_\ell$, it holds that

$$w_k < x < w_{k-1} \implies g(x) = \begin{cases} h_k & \text{for } w_{k+1} \le x < w_k \\ h_K & \text{for } 0 \le x < w_K. \end{cases}$$

Write $\mathsf{B} = \{w_k : 1 \le k \le K\}$ for the set of breakpoints of g.

Under the assumption that f is decreasing, we see that as soon as $g \ge f$ on B, it follows that $g \ge f$ on the full interval $[0,1]$, leaving us to only check for domination at a finite set of ordinates. Moreover, elementary considerations indicate that one 'may as well' enforce equality at the jump points, i.e. that $g = f$ exactly on B.

Constructing Ziggurat Envelopes

So, how shall we find our g? One useful perspective is to forget for a moment the number of slabs K, and to instead think in terms of the area parameter $\alpha \in (0,1)$, the common area of each slab in our ziggurat. From this perspective, we can think algorithmically:

First, lay down a rectangular slab of area α across the whole interval $[0,1]$, yielding $h_1 = \theta_1 = \alpha$. Since $f(1) = 0$, we certainly have domination at $x = 1$. Since $f(0) = 1 > \alpha$ and f is continuous and strictly decreasing, we can find a unique $w_2 \in [0, w_1)$ for which $f(w_2) = \alpha$ (namely, $w_2 = f^{-1}(\alpha)$). Moreover, since f is strictly decreasing, it holds that $f(x) \le h_1$ for all $x \in [w_2, w_1)$, and this interval of domination cannot be extended any further to the left.

[1] Per Wikipedia, 'A ziggurat [...] is a type of massive structure built in ancient Mesopotamia and Iran. It has the form of a terraced compound of successively receding stories or levels.'

To lay down our second slab, the constant-area constraint forces us to take $\theta_2 = \alpha \cdot w_2^{-1}$. Our job is then once more to find the maximal interval on which this yields a valid upper bound, i.e. to identify a minimal $w_3 \in [0, w_2)$ for which $f(x) \leq h_2$ on the interval $[w_3, w_2)$. One possibility is that $h_2 \geq 1$, in which case we are already successfully dominating f on the full interval $[0, 1]$, and are done. This can happen in two ways: either $h_2 = 1$, and we can conclude immediately, or $h_2 > 1$ already, and our slab is thicker than needed. In the latter case, we correct its value to $\theta_2 = 1 - h_1$ and conclude similarly. On the other hand, if $h_2 < 1$, then the same argument as in the previous step yields that we can take $w_3 = f^{-1}(h_2)$ and carry on.

The remainder of the procedure is then analogous: given then $\{\theta_j\}$, $\{h_j\}$, $\{w_j\}$ which have been computed so far, take $\theta_k = \alpha \cdot w_k^{-1}$, and either observe that $h_k \geq 1$ (in which case we terminate, potentially with adjustment to the final thickness), or that $h_k < 1$ (in which case we set $w_{k+1} = f^{-1}(h_k)$, and continue).

Written purely in terms of the height variables, we see that one only has to compute

$$\begin{aligned} & h_1 = \alpha \\ 1 < k \leq K(\alpha), \qquad & h_k = h_{k-1} + \frac{\alpha}{f^{-1}(h_{k-1})}, \end{aligned}$$

where $K(\alpha)$ is minimal such that $h_k \geq 1$. If $h_{K(\alpha)} = 1$ exactly, then we simply use our output as-is to infer $\theta_k = h_k - h_{k-1}$, $w_k = \alpha \cdot \theta_k^{-1}$, and hence reconstruct the ziggurat representation of g. If we overshoot and $h_{K(\alpha)} > 1$, then one makes some light alterations to the final values of (θ_k, w_k, h_k) as needed and concludes similarly.

Algorithm 2 Rejection Sampling from a Ziggurat Representation

1. Sample $I \sim \mathsf{Categorical}((\theta_1, \theta_2, \cdots, \theta_K))$.
2. Given $I = k$, sample $X \sim \mathsf{Uniform}[0, w_k]$ and compute $H = h_k$.
3. With probability $\frac{f(X)}{H} \in [0, 1]$, accept the sample; otherwise, go to Step 1.

From Fixed-Area to Fixed-Budget

Treating α as fixed is useful for the above derivations, but is somewhat frustrating from the point of view of pure computational efficiency. As discussed earlier, there are genuine benefits to working with a number of slabs K which is a power of 2. Moreover, for the final ziggurat representation to exactly satisfy the fixed-area property, we require that $h_K = 1$ exactly. As such, we have to work a little harder.

Fortunately, the problem is sufficiently well-behaved that a numerical solution is not so hard to find. For each $k \in \mathbf{N}$, the mapping $\alpha \mapsto h_k(\alpha)$ is continuous and monotonically increasing, and so solving $h_K(\alpha) = 1$ for a given $K = 2^B$ can be solved by fairly simple search procedures (known in this context as 'shooting' methods) with reasonable theoretical guarantees.

Some Large-K Asymptotics by ODE Approximation

For most efficient sampling, it is of interest to take rather large K or B, and hence rather small α (e.g. for the normal distribution, standard ziggurat implementations take $K = 2^B$, $B \in \{6, 7, 8\}$). Here, an interesting approximation presents itself: the recursions for h_k behave much like an explicit Euler scheme for solving the ordinary differential equation

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{1}{f^{-1}(h)}, \qquad h(0) = 0.$$

Our question then translates to asking when this differential equation passes the level $h = 1$. Writing then

$$f^{-1}(h) \frac{\mathrm{d}h}{\mathrm{d}t} = 1$$

and defining g to be the primitive of the mapping $h \mapsto f^{-1}(h)$, i.e.

$$g(h) = \int_0^h f^{-1}(\eta)\, \mathrm{d}\eta,$$

one checks that the ODE has solution $h(t) = g^{-1}(t)$, and hence crosses $h = 1$ at

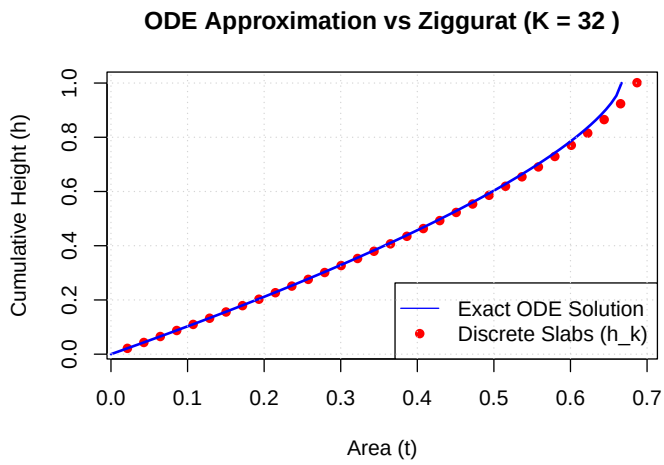
$$t_\star = g(1) = \int_0^1 f^{-1}(\eta)\, \mathrm{d}\eta = \int_0^1 f(x)\, \mathrm{d}x.$$

(The last equality is much easier to prove visually than algebraically.)

As a result, in order to obtain a ziggurat representation with approximately K slabs, it is reasonable to take an initial guess α_0 such that $K\alpha_0 = \int_0^1 f(x)\, \mathrm{d}x$, and then refine this initial guess by numerical methods.

With the benefit of hindsight, there is another interpretation of this approach, which is to find heights such that the exact integrals $\int_{h_{k-1}}^{h_k} f^{-1}(\eta)\, \mathrm{d}\eta$ are of constant size, so that their rectangular approximations might also be of constant size.

Figure 2: The ODE approximation with $K = 32$ slabs



Loosening the Assumptions

The choice to work on the unit square $[0,1]^2$ is entirely cosmetic, and simplifies the initial derivations, but is certainly inessential.

The assumption that the density be continuous and strictly decreasing is slightly stronger than needed, and everything still goes through for densities which are only decreasing, albeit with some extra checks and balances to account for well-definedness of different quantities.

The assumption that the density have compact support can also be dispensed with if the upper tail of the density is sufficiently well-understood. Roughly speaking, the initial slab shall be taken as a wide, thin rectangular slab, with the tail of the density adjoined to its right. Having designed this initial slab, the remainder of the density is again of compact support, and the main algorithm applies without modification.

Fast Acceptance Steps

A particular practical benefit of the ziggurat approach is the possibility to frequently accept proposals without having to evaluate the target f. Indeed, if we propose X from the kth slab $\mathbf{1}[0 \le x < w_k]$ and observe that $X < w_{k+1}$, then we know with certainty that the proposal will be accepted (in effect, we have generated X

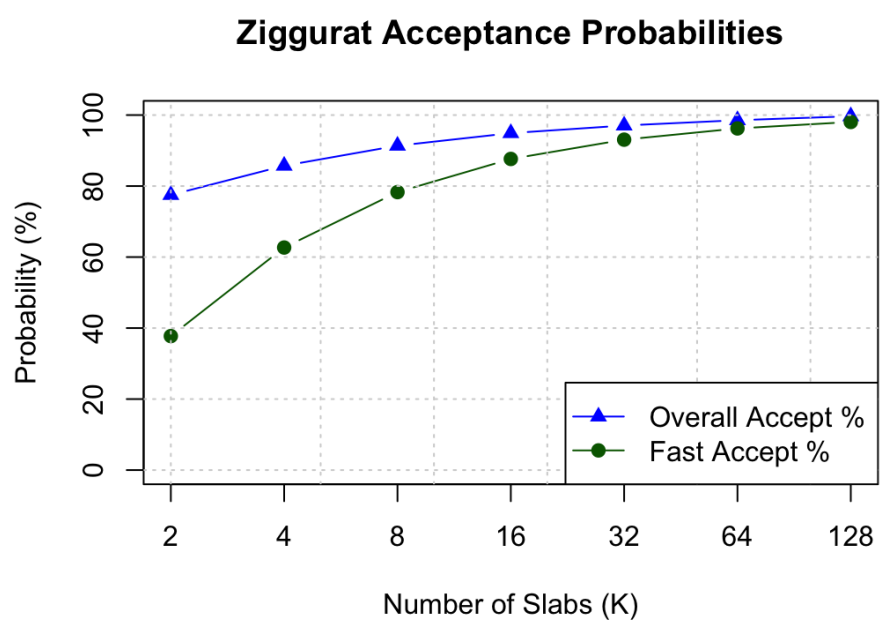
from the part of the ziggurat which 'clearly' lies beneath the curve $y = f(x)$), and so one doesn't need to evaluate $f(X)$ (which is typically the more expensive part of a rejection sampling routine).

Algorithm 3 Rejection Sampling from a Ziggurat Representation with Fast Acceptance

1. Sample $I \sim \mathsf{Categorical}\left((\theta_1, \theta_2, \cdots, \theta_K)\right)$.
2. Given $I = k$, sample $X \sim \mathsf{Uniform}\left[0, w_k\right]$ and compute $H = h_k$.
   (a) If $X < w_{k+1}$, then accept the sample immediately.
   (b) Otherwise, with probability $\frac{f(X)}{H} \in [0, 1]$, accept the sample
   (c) Otherwise, go to Step 1.

Moreover, as K grows, far from being an 'added bonus', this is instead the typical behaviour. Figure 3 demonstrates that for $K = 16$, this takes place for around 87% of proposals, and with $K = 128$, the number is upwards of 98%. Note that this is a lower bound of the total acceptance probability.

Figure 3: Overall and fast acceptance probabilities for ziggurat method.



Conclusion

The Ziggurat algorithm is an extremely efficient approach to simulating from structured univariate distributions which will be called upon repeatedly. Its preparation requires the solution of an interesting univariate optimisation problem, which can be solved quite easily in practice, and leads to implementations which are very hardware-friendly, behaving much like a discrete lookup table much of the time.